Statistical vs. Machine learning

N'GUESSAN Maeva, DUTAUZIET Roland, ASSE Bethuel

11th February 2023

Abstract

Whether in statistical learning or machine learning, it is essential to select the relevant features. This report therefore focuses on the presentation, evaluation, and comparison of procedures that can be used for this selection. It is based on selection methods specific to statistical learning (procedures with inference) and others specific to machine learning (procedures without inference). It presents these different techniques tested on simulated data, according to various stopping criteria or conditions. In our data generation, we consider different scenarios: independence (null hypothesis), dependence (alternative hypothesis), including situations of linear dependence, structural break, and multi-colinearity. Outliers are also taken into account.

This paper also aims to provide a detailed analysis of feature selection methods in statistical learning and machine learning, to better understand the strengths and weaknesses of each approach. This will also help to guide experts in their choice of methods best fitted to their specific needs. This study reveals the key outcomes that follow : machine learning selection methods generally surpass statistical learning methods in terms of performance. Nevertheless, the performance of these methods is compromised when outliers are present, and in a few rare cases when structural breaks are introduced. This study also shows that machine learning processes are much more affected when coefficients levels vary. This also happens when the noise inserted in the model moves. This susceptibility to variations is most apparent when there is correlation.

Keywords : Machine Learning, Statistical Learning, Feature selection, Lasso, Lars, Elasticnet, Forward, Backward, Stepwise, Bootstrap, K-Fold, Cross-Validation.

Directed by Phillipe DE PERETTI

UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

Contents

1 Introduction

Statistical and machine learning are fields that have evolved significantly since their appearance in the mid-twentieth century. Historically, statistical learning has its origins in the work of pioneers such as Ronald Fisher and Karl Pearson, who set the foundations for statistical modeling with ANOVA, the Fisher test, the chi-squared test, and more. Over the decades, this discipline has developed by integrating concepts of probability and statistical inference. At the same time, statistical learning emerged with the rise of the computer era, driven by the need to create computer programs that could learn from data. The term "Machine Learning" was popularized by Arthur Samuel in 1959. Other pioneers in this field include names like Geoffrey Hinton and Andrew Ng, known for their work on deep learning and neural networks. Technological advances and the growing accessibility of data have driven this field ahead, giving it increasingly advanced methodologies. The choice of this topic comes from the increasing role of these two approaches in the understanding and analysis of large quantities of data, mainly in the healthcare, finance, and technology areas.

These two notions are often confused because they are wrongly considered to be the same thing. Although the difference between these two approaches is not necessarily obvious, it does exist and lies mainly in their purpose. While statistical learning focuses on understanding the relationships between variables based on inferential models, machine learning targets the optimization of predictive performance without relying on statistical laws. In this paper, we review these various methodologies, focusing on automatic variable selection, a key aspect in the analysis of databases containing a multitude of possible predictors.

this paper aims to measure and compare the performance of automatic variable selection between these two techniques. More exactly, we will introduce several variable selection methods in statistical learning and then in machine learning, and test them several times on simulated data. This will help us to estimate their performance using different stopping criteria. These methods will then be compared with each other to observe the differences in results.

We'll be looking in detail at underfitting and overfitting, two critical issues in predictive modeling. Underfitting happens when the model is too simple to catch the complexity of the data, leading to poor performance on both training and test data. On the opposite, overfitting occurs when a model fits the training data too precisely, capturing its noise to the point of being inefficient in prediction on test data. This often happens with models that have too many parameters compared to the size of the training data. This work aims to determine how these phenomena affect the variable selection and consequently model performance.

This being said, our study will unfold in several stages: in section 2, we will present the different selection methods and their criteria according to our 2 approaches. Section 3 will highlight the model evaluation metrics and data generation processes. In Part 4, we present the results of the different data generation processes. Finally, Part 5 will focus on an empirical analysis carried out on real data.

2 Theoretical Approach

2.1 Introduction to Feature Selection Algorithms: Forward, Backward and Stepwise

2.1.1 Forward Method

Forward stepwise selection is a strategic approach in statistical modeling that begins with a null model, which includes no variables. The method focuses on sequentially adding the most significant variables to the model. The significance of each variable is determined by criteria such as its p-value, contribution to R^2 increase, or its effect in reducing the model's Residual Sum of Squares (RSS). This process continues until it meets a pre-established stopping rule or all variables under consideration are incorporated into the model. The stopping rule is often based on a threshold for the p-value, which can be a fixed value or determined by statistical measures of parsimony that weigh the complexity of the model against its fit to the data, taking into account both the number of explanatory variables and the sample size. [16]

2.1.2 Backward Method

Backward selection, also known as backward elimination, is a statistical method for variable selection that begins with a full model containing all potential variables. It involves sequentially removing the least significant variables, identified by the highest p-value, the smallest decrease in R^2, or the smallest increase in RSS (Residual Sum of Squares). This process continues until a pre-set stopping rule is met, typically when all remaining variables in the model have p-values below a specified threshold, which can be a fixed value or determined by parsimony criteria. The method concludes by returning a simplified model containing only the statistically significant variables. [16]

2.1.3 Stepwise Method

The stepwise method is a variable selection that combines the approaches of the forward and backward methods. In this method, after each addition of a variable to the model (as in the forward method), all variables already present in the model are reassessed to determine if their significance has changed, usually using a specific tolerance threshold.

If a variable becomes insignificant after the addition of a new variable, this indicates a form of correlation between the variables, a phenomenon that is not detected by either the forward or backward methods alone. In this case, the stepwise method follows the process of the backward method: it eliminates the least significant variable, even during the same iteration in which a new variable has been added.

This approach allows for consideration of complex interactions between variables, as the addition or removal of a variable can affect the significance of others. The stepwise process continues until the addition or removal of an additional variable no longer contributes to the optimization of the stopping criterion. [16]

2.2 Statistical criteria for features selection

In statistical learning, feature selection criteria are based on the theoretical distribution of a population. These criteria assess the significance of features in a model relative to the model's underlying statistical laws. This approach ensures that the selected features are not only relevant but also statistically significant within the theoretical framework of the model.

2.2.1 T-Test

Introduced by William Sealy Gosset under the pseudonym "Student" in 1908[6], the T-test has become an essential tool in statistics, particularly in biomedical and social research fields. In the context of variable selection, the T-test is commonly used to determine the statistical significance of individual predictors within a model. Under the assumption that the mean of the sample follows a Normal distribution, we have:

$$t_{\hat{b}} = \frac{\hat{b}}{\hat{\sigma}_{\hat{b}}} \sim T(N - (P + 1))$$

With N the size of the sample, P the number of features, and $\sigma_{\hat{b}}$ the estimated standard error of the model. Under the null hypothesis (typically, the hypothesis that there is no effect or no difference), the t-statistic tends asymptotically towards a Normal distribution, as the sample size (N) becomes large. The p-value, representing the probability of observing a test statistic as extreme as the observed one under the null hypothesis(H0), is compared to α (the pre-set error threshold); if the p-value is less than α, the feature is considered statistically significant.

2.2.2 Adjusted $\mathbf{R^2}$

The R-squared is a measure of the proportion of variance in the dependent variable that is predictable from the independent variables. However, It has a known limitation: it always increases as more predictors are added to the model, even if those predictors are not statistically significant. This can lead to overfitting if the model complexity is not taken into consideration. The adjusted R-squared corrects[12] this by incorporating the number of predictors and the sample size into its calculation, thus penalizing the addition of non-significant features. The formula for adjusted R-squared is:

$$\text{Adjusted } R^2 = 1 - \left[\frac{(1 - R^2)(N - 1)}{N - (P + 1)}\right]$$

Where :
$R^2 = 1 - \frac{RSS}{TSS}$: RSS= Residual Sum of Squares ; TSS= Total Sum of Squares
N is the number of observations
P is the number of predictors

2.2.3 F-Test

The F-test, proposed by Fisher in 1935[6], is a statistical method applied to evaluate the collective significance of all features in a model. Suitable for finite-sample scenarios, it operates under the hypothesis that samples are independently and randomly selected (i.i.d) from a Normal distribution, with each being independent and having the same variance. The test compares the null hypothesis:non-significance of all the features (except the intercept) against the alternative hypothesis: significance of at least one of the coefficients of the full model $Y = X\beta + \varepsilon$ (with X the feature matrix, β the coefficient's vector and ε the noise). So, we have:

$$F = \frac{\frac{RSS_c - RSS}{P}}{\frac{RSS}{N-(P+1)}} \sim F(P, N-(P+1))$$

With $RSS = \sum_{i=1}^{N}(Y - X\hat{\beta})^2$ and RSSc as the RSS under H_0. Then, we can use either Fisher's distribution or the p-value to determine the rejection area. This test is less commonly used because, while it can confirm the presence of at least one non-zero coefficient, it fails to identify the specific coefficients involved. This makes it less precise compared to the T-test.

2.2.4 Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation (MLE) is a method for estimating the parameters from their theoretical distribution, introduced in the early 20th century by Fisher[**fisher˙absolute˙1997**]. It consists in finding the parameter values that maximize the likelihood function, which is the probability of observing the given data under a specific statistical model. With θ the vector of parameters of the theoretical distribution of $Y|X = x$. and $\hat{\theta}$ its estimation we have:

$$L(\theta \mid \begin{pmatrix} x(1) \\ y(1) \end{pmatrix}, \ldots, \begin{pmatrix} x(N) \\ y(N) \end{pmatrix} = \prod_{n=1}^{N} f_\theta\left(x^{(i)}\right)$$

We get:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\ L_\theta(x_1, \ldots, x_n)$$

With f as the theoretical density function of $Y \mid X = x(i)$.
The MLE is consistent (the estimate parameter converges to its true value as the sample size increases), asymptotically normal, and asymptotically efficient (it achieves the lowest possible variance among all consistent estimators). This approach is based on theoretical statistical laws and offers a statistical criterion for comparing different sets of parameter vectors.

2.2.5 The likelihood Ratio Test

Under the principle of MLE previously explained, this test [13] compares the goodness of fit of two statistical models. It's useful in feature selection for comparing a constrained model (with fewer features) against a full model (with all features), to determine if the reduced model is sufficient. The hypotheses are set as :

$$\begin{cases} H_0 : \theta \in \Theta_0 \\ \text{contre} \\ H_1 : \theta \in \Theta_1 \supset \Theta_0 \end{cases}$$

We define $T(X_1, \ldots, X_N)$ as test-statistic:

$$T = -2 \log \left(\frac{L(\hat{\theta}_0 \mid \begin{pmatrix} x(1) \\ y(1) \end{pmatrix}, \ldots, \begin{pmatrix} x(N) \\ y(N) \end{pmatrix})}{L(\hat{\theta} \mid \begin{pmatrix} x(1) \\ y(1) \end{pmatrix}, \ldots, \begin{pmatrix} x(N) \\ y(N) \end{pmatrix})} \right) \sim \chi^2(\dim(\Theta) - \dim(\Theta_0))$$

And a rejection region $W = T > C$. We reject H0 if $T(X_1, \ldots, X_N)$ belongs to W. C is calculated according to α.

In feature selection, the test evaluates the information content between a full model and a reduced model. If the reduced model loses too much information compared to the full model, the test will reject it. However, the decision to reject the null hypothesis is significantly influenced by the chosen α value. We generally choose $\alpha = 5\%$ but the choice of its value can lead to different results in the model selection.

2.2.6 Akaike Information Criterion

The Akaike Information Criterion (AIC) was introduced by Hirotugu Akaike in 1973 [1] and presented the AIC as a new method for model selection in statistics, offering a criterion based on information theory principles. It estimates the relative quality of statistical models for a given dataset by balancing the model's complexity against how well the model fits the data. The formula for the AIC is:

$$\text{AIC} = 2p - 2\ln(\hat{L})$$

Where : p is the number of parameters in the statistical model: So it penalizes more complex models and prevents overfitting. $\hat{L}$ is the likelihood of the model: representing the fit of the model to the data.

The goal of model selection using AIC is to find the model with the lowest AIC value. A lower AIC value indicates a model with better parsimony (balance between the goodness of $fit(-2ln(L))$ and simplicity(2k)). Additionally, the effectiveness of the AIC depends on the assumption that the maximum likelihood estimators converge to a Normal distribution in large samples (asymptotic normality). Especially in the linear Gaussian model, we have :

$$AIC = 2p + N\left[\ln\left(\frac{2\pi \times RSS}{N}\right) + 1\right]$$

2.2.7 Corrected Akaike Information Criterion (AICc)

For small sample sizes, AICc provides a more accurate model selection criterion than AIC [2]. We have:

$$AICc = AIC + 2\frac{p(p+1)}{N-p-1}$$

AICc converges to AIC as long as N is very large or p is very small, making AICc more suitable for smaller N sizes or bigger p sizes.

2.2.8 Schwarz Bayesian Criterion(SBC)

Similar to AIC, the Schwarz Bayesian Criterion (SBC) is used for model selection, but it imposes a stricter penalty for the number of parameters in the model. It can be more appropriate for larger datasets. In his 1978's paper [15], Schwarz developed the SBC as an asymptotic approximation to a transformation of the Bayesian posterior probability of a model, providing a criterion for model selection. The SBC is given by:

$$SBC = -2\ln(\hat{L}) + p\ln(N)$$

When N is larger, the SBC will disqualify overly complex models. However, with a small sample, the SBC criterion tends to underfit a model. In other words, SBC-selected models tend to be less complex than the AIC ones, but generally with less goodness of fit.

2.2.9 Sawa's Bayesian Information Criterion (BIC)

Sawa's BIC (BIC) is a variant of the traditional BIC [14] that might be more appropriate in certain contexts. The difference lies in the penalty term for the number of parameters to adjust for finite sample sizes and potential overfitting. It is designed to perform better in small sample situations by providing a more accurate penalty term.

BIC is a function of the number of observations N, the SSE, the pure error variance fitting the full model, and the number of independent variables including the intercept.

$$BIC = N * \ln\left(\frac{RSS}{N}\right) + 2(p+2)q - 2q^2$$

Where $q = N * \frac{\sigma^2}{RSS}$ and p is the number of model parameters including intercept. However, "Sawa's BIC" is not standard in the statistical literature and is often confused with the SBC.

2.3 Machine Learning Processes and Selection

2.3.1 Introduction

Machine learning is training computers to learn from data and make decisions or predictions based on that learning. An essential concept in machine learning and particularly in feature selection is the minimization of the loss function. A loss function is the measure of the error between the predictions made by machine learning algorithms and the actual data. There are various types of loss functions. One of the earliest and most commonly used is the Mean Squared Error (MSE) [**lawless1981mean**]; which is derived from the method of least squares formalized by Adrien-Marie Legendre and Carl Friedrich Gauss in the early 19th century. The MSE is:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - (\alpha + \beta^T X_i))^2$$

Within the context of feature selection, the machine learning processes use a penalization norm where λ is a control parameter to be determined. In this section, we will deal with these different machine learning processes and then study the theoretical impact of the choice of λ value.

2.3.2 The Lambda Penalty

The lambda penalty refers to regularization added to a cost function to prevent overfitting. Regularization introduces a penalty term to the model's loss function, encouraging simpler models. Thus, the lambda penalty controls the strength of this regularization term. Indeed, the higher its value, the stronger the regularization effect. How, then do we choose the optimal value for λ? We will explore this question using explanatory criteria, such as the minimization of the Mallows Cp or predictive criteria, such as the k-fold cross-validation.

Common forms of regularization include the L1 penalty (Lasso) and the L2 penalty (Ridge).

2.3.3 Least Absolute Shrinkage and Selection Operator: LASSO

LASSO, introduced by Robert Tibshirani in 1996 [17] is a regression method that performs both variable selection and regularization to enhance the accuracy and interpretability of the model. The penalized loss function of LASSO is given by:

$$MSE_{\lambda}^{LASSO}(\theta) = \frac{1}{2N}\sum_{i=1}^{N}\left(Y_i - (\alpha + \beta^T X_i)\right)^2 + \lambda\|\theta\|_1$$

Where : $\|\theta\|_1 = \sum_{j=1}^{p} |\beta_j|$

Thus, LASSO constrains the sum of the absolute values of the coefficients, allowing it to contract (shrink) some coefficients to zero, and highlight those with the most impact. L1 regularization produces more interpretable models and can handle problems where the number of variables is greater than the number of observations ($N < p$). However, the choice of λ is crucial and typically done through cross-validation. Moreover, LASSO might not perform as well when variables are highly correlated; it will tend to select only one feature among a group of highly correlated features and shrink the other ones to zero even though they have an impact on the dependent feature. That's the "compressed sensing problem"[19].

2.3.4 Ridge Penalty

The concept of Ridge regression, also known as L2 regularization, was proposed by Arthur E. Hoerl and Robert W. Kennard in 1970 [10]. In simple terms, It is similar to LASSO but uses the sum of the squares of the parameters for regularization. The Ridge penalized MSE is:

$$MSE_{\lambda}^{\text{Ridge}}(\theta) = \frac{1}{2N}\sum_{i=1}^{n}\left(Y_i - (\alpha + \beta^T X_i)\right)^2 + \lambda\|\theta\|^2$$

Where the Euclidean norm : $\|\theta\|^2 = \sum_{j=1}^{p}\beta_j^2$

This technique fixes issues of multicollinearity in linear regression models by adding a degree of bias to the regression estimates. Ridge helps to reduce model complexity and improves its interpretability while preventing overfitting. It's especially useful when all features are potentially relevant. However, it also depends on λ, which requires cross-validation. But for a better performance in the case of multicollinearity, a mixture of Ridge and LASSO regression is preferable, which is the Elastic Net regression.

2.3.5 Elastic Net methods

The Elastic Net is a regularization method that combines the penalties of LASSO (L1) and RIDGE (L2), The objective of this regression is therefore twofold: to make the least relevant features tend towards 0 as well as to standardize the value of the remaining features as illustrated by the following equation

$$J(W) = \frac{1}{2N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \lambda\sum_{j=1}^{p}\alpha$$

In this equation, J(W) represents the cost function that is minimized during the training of the model. The first term is the mean squared error of the predictions $\hat{y}_i$ against the actual values yi. The second term is the Elastic Net penalty, which is composed of two parts: the first is the L1 penalty (the sum of the absolute values of the coefficients $\sum|\beta_j|$, and the second is the L2 penalty (the sum of the squares of the coefficient $\sum\beta_j^2$)

The parameter α determines the balance between the L1 and L2 penalties. An α of 0 implies that only RIDGE is used, while an α of 1 implies that only LASSO is used. The parameter λ controls the overall strength of the penalty. To select the best parameters α and λ, a grid search (hypergrid) is typically employed, which evaluates multiple combinations of these parameters and selects the one that offers the best performance according to a predefined criterion.

The advantage of Elastic Net is that it can more effectively handle correlated predictors compared to the LASSO method alone. The use of the L2 norm limits the excessive shrinkage of the coefficients, which is typical of the L1 norm. However, the success of the Elastic Net method depends heavily on the proper selection of the parameters α and λ, and improper tuning of two parameters can lead to poor model performance [20].

2.3.6 Incremental Forward Stagewise: IFS

Incremental Forward Stagewise (IFS), also known as the Slow learning algorithm, offers an alternative approach to linear regression akin to boosting [7]. This method diverges from traditional stepwise algorithms, which typically add or remove features in entirety; IFS instead incrementally adjusts a selected coefficient by a small quantity at each iteration, ensuring a more granular refinement of the model. The IFS algorithm commences by defining a learning rate ϵ and initiating with the residual $r^{(0)} = y - \hat{y}$, where all coefficients of regression are initially set to zero. The algorithm proceeds by identifying the feature $x(j)$ that shares the highest correlation with the residual $r^{(k)}$. Subsequently, the coefficient β_j is updated by $\beta_j^{(k)} = \beta_j^{(k-1)} + \epsilon \cdot \mathrm{sign}(x_j^T, r)$, and the residual $r^{(k)} = y - X\beta^{(k)}$.

These steps are iteratively executed until the residuals are no longer correlated with any feature or until the residuals are no longer correlated with any feature or until a predetermined criterion, such as the Akaike Information Criterion (AIC), satisfied.

The IFS algorithm often leads to selections like those made by the Lasso method, yet it is distinguished by its less aggressive selection strategy. For instance, while the Lasso path may exhibit instability due to non-monotonic behavior, especially in the presence of correlated features, the IFS pathway maintains stability.

2.3.7 Least Angle Regression Shrinkage: LARS

The Least Angle Regression Shrinkage (LARS) technique, introduced by Bradley Efron, Trevor Hastie, Iain Johnstone, and Robert Tibshirani in 2004[4], presents an innovative approach to linear regression modeling, deviating from traditional loss function-based methods. Instead, LARS utilizes a geometric approach to integrate predictors into the model. This process begins with standardizing features to have zero mean and unit variance, and commences with a residual $r_i = y - \hat{y}$ with an initial coefficient $\beta_i = 0$.

LARS's methodology involves iteratively adding predictors to the model. The selection of these predictors is based on their correlation with the residual. The algorithm continuously adjusts the coefficient of the most correlated feature, increasing it towards its least-squares coefficient. This adjustment continues until another feature exhibits an equal correlation with the residual. The process then extends to include this second feature, adjusting their coefficients together until a third feature shows higher correlation with the residual. This sequence repeats until all features are incorporated into the model.

A notable aspect of LARS is its similarity to, yet distinct differences from, the LASSO (Least Absolute Shrinkage and Selection Operator) technique and IFS. With minor modifications, LARS can replicate the LASSO's solution set, but it stands out due to its lower sensitivity to noise, which enhances its robustness and stability. Unlike LASSO, which imposes zero constraints on some regression coefficients, LARS does not enforce such constraints. Instead, it opts for a dynamic inclusion or exclusion of features in the model. This approach can be particularly advantageous in situations with correlated features, potentially reducing the risk of overfitting that LASSO might encounter under similar circumstances.

2.3.8 Mallow's CP

Mallow's Cp is a statistical measure used to assess the fit of regression models in statistics. Developed by Colin Mallows in 1973 [11], "Criterion p"(Cp) aims to identify models that achieve a balance between complexity (in terms of the number of predictors) and the fit of the model to the data. It is particularly useful for comparing regression models that have different numbers of independent variables. The formula for Mallows' Cp is:

$$\frac{SSE_{k^*}}{\sigma^2} + 2k^* - N$$

Where: SSE_{k^*} is the sum of squared errors for the model with $k*$ parameters, σ^2 is an estimated variance of the full model, $k* < k$ represents a subset of parameters in the model(including the intercept). The goal is to find a model with a Cp value near its number of predictors.

Note that Mallow's Cp is an explanatory criterion. It means that it follows the parsimony principle that is the trade-off between fitting the data well (which usually requires more parameters) and keeping the model as simple as possible (to avoid overfitting).

2.3.9 Predictive criteria : K-Fold and Press

Another type of criteria are the predictive ones. Those criteria involve partitioning a dataset into a training set and a test set. We fit the model on the train set and then make predictions on the test set. A predictive criterion selects the model with the best performance in the test set.

The K-fold validation(or cross-validation), was first introduced by Mosteller and Tukey in 1968 and further developed by Geisser in 1975 [5]. It divides the dataset in K approximately equal-sized subsets. It selects a subset among them a validation set and trains the model on the remaining (K-1) subsets. This process is repeated K times, with each of the k subsamples used exactly once as the validation data. The chosen model is the one that minimizes the average errors across all k validation sets, generally using the MSE.

When $K = N$, it is a special case called "leave one out" or PRESS. The PRESS statistic was proposed by Allen in 1974 [3]. It's the same principle as K-fold but with the maximum value of K. Hypothetically, as the number of folds increases, the bias in the error estimate on the test set is expected to decrease, but may also lead to higher variance in the estimate due to the similarity of when K is large. The PRESS statistic is defined as:

$$PRESS = \sum_{i=1}^{N} (y_i - \hat{y}_{i(-i)})^2$$

Where y_i is the actual observed value for the ith observation and $\hat{y}_{i(-i)}$ is the predicted value for the ith observation from the model fitted without the ith observation. A lower PRESS value indicates a model with better predictive ability.

3 Data Generating Process

To test our feature selection techniques, we will first generate data on which we will apply these techniques. Then, we will construct specific metrics to evaluate and compare their performance.

3.1 Metrics used for model evaluation

Metrics are important for analyzing and comparing feature selection models. In our approach, each metric acts like a cumulative indicator, incrementing by 1 each time a specific condition is satisfied. At the end of all iterations, we compute probabilities by dividing the cumulative value of each metric by the total number of iterations. For our analysis, we have set up five diverse metrics : perfect fitting, apparently perfect fitting, good overfitting, bad overfitting, underfitting.

Metrics are used to compare the real features of the model with the features selected by the selection method applied. When the features selected by the model are exactly the same as the right features, the perfect fitting condition is filled. In another situation, where the method used seems to select the right numbers of features, but not necessarily the right ones, the "apparently perfect fitting" condition is also satisfied. The cases of overfitting and underfitting depend on the number of features selected by the algorithm. Overfitting happens when the algorithm selects far more features than the exact number features. There are two forms of overfitting :

- Good overfitting : in this case, the algorithm selects all the correct features, but also adds other irrelevant features. This means that it correctly captures the essential features but goes beyond them, including unnecessary information.

- bad overfitting : here, the algorithm selects more features than required, but misses some of the truly important ones. This leads to a model that overfits the specific training data, including wrong or irrelevant features, while missing the right ones.

Underfitting is when the algorithm selects fewer features than the right number required. Thanks to these 5 metrics, we can estimate the probability that the selection method will result in underfitting, overfitting, perfect fitting, and so on.

Metrics in this context work like quantitative indicators. For each iteration of a feature selection algorithm, the metrics act as a counter that increases by 1 each time a specific condition is met. For example, if an algorithm is perfectly successful in selecting the right features, the metric for "perfect fitting" increases by 1. After many iterations, these counters can be used to determine the probability or frequency with which each condition (such as perfect fitting, underfitting, etc.) is met, by dividing the cumulative total of each counter by the total number of iterations. This provides a quantitative measure of the performance of feature selection techniques.

3.2 Data Generating Process with Independence (White noise)

For data generation with white noise, our approach is based on the use of a multivariate normal distribution in which we have 100 features and 100 observations. The second step consists of initializing the mean to 0 and setting an identity matrix of size equal to the number of features, in our case 100. This identity matrix corresponds to the variance-covariance matrix. It indicates that each feature has a variance of 1 and that there are no correlations between them. Having set up the 100 features, we build a model using 6 of the 100 features. The zero mean of the multivariate normal distribution also means that each feature in the model has an expected mean of 0. Let's note these X1 to X6. Each of these 6 features has a specific coefficient associated with it, and we note β the column vector of these 6 coefficients. Finally, we add ϵ drawn from a normal distribution. This last step completes the final model, which takes the form of a linear equation in which each feature Xi is multiplied by its respective coefficient, with the addition of noise ϵ to represent error or unexplained variation. In our simulation context, coefficients are arbitrarily defined to test the model. Our model is therefore as follows:

$$Y = X_1 + 2.1X_2 + 0.6X_3 + 2X_4 + 1.5X_5 + 2.9X_6 + \epsilon$$

To introduce a moderate level of noise, slightly affecting the predicted values without completely dominating the effect of the independent features, we multiply the noise ϵ by a scaling factor equal to 0.1. This value is appropriate if we wish to adjust the impact of noise on model results without impacting the actual properties of the noise.

After generating Y from features X1 to X6 and adding noise, we concatenate Y with all the features to get a complete dataset. With the dataset ready, we can now test the feature selection methods. To analyze our results and evaluate the efficiency of each selection technique by using our metrics, this procedure is iterated 1000 times, including these steps :

1. Generation of uncorrelated data with white noise
2. Calculation of Y
3. Generation of a dataset including Y and the features
4. Regression of Y on features X_1 to X_50. This is where one of the feature selection methods is used and it is by this method that the regression is made.
5. The process starts again from the first step.

3.3 Data Generating Process with Outliers

Outliers in distribution are observations that are significantly different from the other values in the dataset and may result from measurement errors, data entry mistakes, or true unusual variations within the population. They can distort statistical analyses, leading to biased parameter estimates and invalid inferences if not appropriately addressed. To add these outliers, we will generate two groups of data: The three first features: X_1, X_2, and X_3 with outliers as the first group and X_4,X_5 and X_6 without outliers as the second one. Generating outliers on these features will allow us to analyze their effects by comparing these groups.

To generate our first group, we use two types of normal distribution: they both generate uncorrelated observations but with different means: 0 or 3. To decide which distribution we use for each observation, we draw a number from a uniform distribution. If this number is greater than 0.2, the standard normal distribution is used, otherwise, we use the normal distribution with a mean of 3.

To generate our second group, we use a standard Normal Distribution. After generating the dataset, we compute the y values and apply a feature selection algorithm for the regression. This entire process is repeated 1000 times to ensure statistical robustness and to compute probabilities using our chosen metrics.

3.4 Data Generating Process with Correlation

Correlation refers to the statistical relationship between two or more variables. We will distinguish two types of correlation: The one we call "intra-correlation" is the correlation within our 6 features, and the "inter-correlation" is the correlation between all the features.

3.4.1 Correlation between the 6 features

To add correlation in our generated data, we have to modify our variance-covariance matrix. It should be positive definite and have non-zero coefficients. A way to do this is to use the Toeplitz matrix, which is a matrix with constant diagonals, with each element along a diagonal being the same. For our 6 features, we wanted to make the correlations decrease as variables become more distant from each other in order. Hence, we have this variance-covariance matrix for the first 6 features:

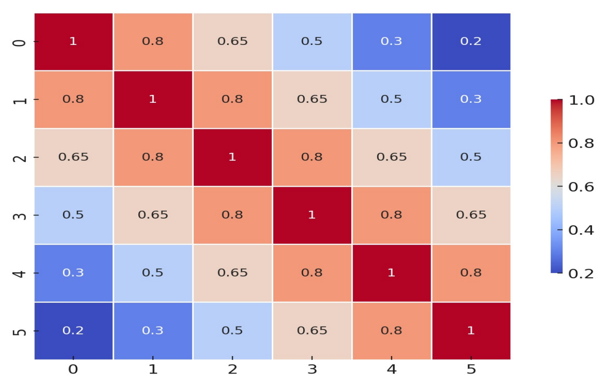


Table 1: Variance Covariance Matrix for the right features

Then, we use an order 94 identity matrix as the variance-covariance matrix for the 94 other features. Finally, we concatenate them into a 100x100 block matrix. This block matrix is then used to generate our observations matrix using a multivariate normal distribution with a mean equal to 0. After this is done, we follow the same steps previously seen and we repeat this process 1000 times to compute probabilities and our metrics.

3.4.2 Correlation between all the features

Similarly, generating data with correlation between all features requires modification of the variance-covariance matrix, which is done using the Toeplitz matrix. This method makes it possible to create correlations between features in a dataset. A vector of 100 elements is built to highlight these correlations, with decreasing intensity as the distance between features increases. In other words, this means that the first features are more strongly correlated with each other than the features that follow in order. For example, X1 is most strongly correlated with X2, the second variable with which X1 is most strongly correlated is X3, but to a lesser degree than with X2, and so on. This $10 * 10$ table is used as an example to explain the method for constructing the full variance-covariance matrix (which is $100 * 100$ in size).

	0	1	2	3	4	5	6	7	8	9
0	1.000	0.619	0.516	0.441	0.384	0.339	0.303	0.274	0.249	0.228
1	0.619	1.000	0.619	0.516	0.441	0.384	0.339	0.303	0.274	0.249
2	0.516	0.619	1.000	0.619	0.516	0.441	0.384	0.339	0.303	0.274
3	0.441	0.516	0.619	1.000	0.619	0.516	0.441	0.384	0.339	0.303
4	0.384	0.441	0.516	0.619	1.000	0.619	0.516	0.441	0.384	0.339
5	0.339	0.384	0.441	0.516	0.619	1.000	0.619	0.516	0.441	0.384
6	0.303	0.339	0.384	0.441	0.516	0.619	1.000	0.619	0.516	0.441
7	0.274	0.303	0.339	0.384	0.441	0.516	0.619	1.000	0.619	0.516
8	0.249	0.274	0.303	0.339	0.384	0.441	0.516	0.619	1.000	0.619
9	0.228	0.249	0.274	0.303	0.339	0.384	0.441	0.516	0.619	1.000

Table 2: Variance Covariance Matrix for all the features (crop of the 10 first features)

After modifying the variance-covariance matrix of all features, we move on to the data generation described in section 3.2. The same procedures are followed after the generation of correlated data, i.e. from the prediction of the value of Y to the regression of Y on the first 50 features. It is at this last stage of regression of Y on the first 50 features that one of the feature selection methods is applied, and it is by this method that the regression is done. Then, in order to analyze the results and test the performance of these selection algorithms, we repeat these steps 1000 times to measure the probabilities using our metrics.

3.5 Data Generating Process with Structural Break

The data generating process with a structural break involves creating a dataset where a significant change occurs at a specific point in time or after a certain number of observations. This change can affect the relationship between variables, the variance within the data, or the mean level of the series. To simulate this scenario, we will process in 2 steps. First, we generate all the observations, using a multivariate Normal Distribution with independence. Subsequently, we add a constant value of +0.5 to each of the 50 last observations. This mimics a scenario with two distinct periods: before and after the break. Once our data is generated, we compute the y value and regress the first 50 features. We repeat these steps 1000 times and we estimate the frequencies, giving insights for our metrics.

4 Results

The results are conditioned by the data generation process, where the optimum stopping criteria are selected from the choice criteria (AIC, AICC, SBC, CP, K-Fold and PRESS). This selection is made following a combination between each selection criterion and each stopping criterion, with the combination giving the best performance being chosen.

4.1 Independence DGP

4.1.1 Statistical Learning: Forward and Backward

In our study, we applied statistical learning techniques, based on four distinct criteria - AIC, AICC, SBC and CP - using Forward and Backward approaches.

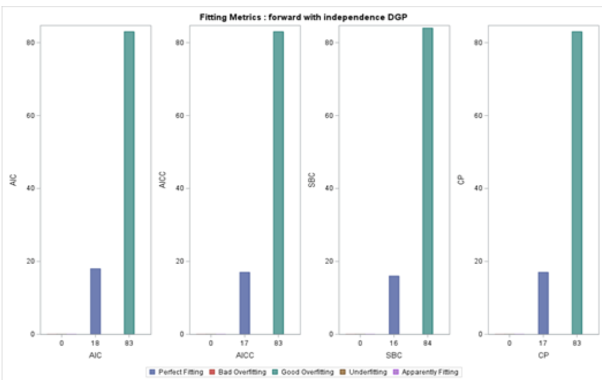


Figure 1: Fitting metrics : Forward in independence

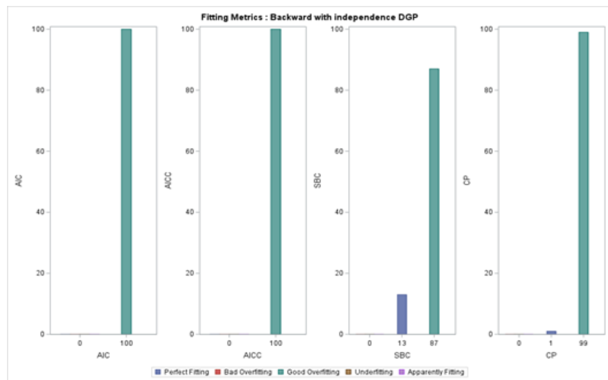


Figure 2: Fitting metrics : Backward in independence

Our tests revealed a significant overfitting effect with the Backward and Forward methods: they properly select the 6 good features X1 to X6, but also add other unexpected features. More exactly, the Forward method led to around 83% overfitting and 17% perfect fitting, whatever the criterion used. The Backward method gave even less satisfying results, with 100% overfitting for the AIC, AICC and CP criteria, only 13% perfect fitting and 87% overfitting for the SBC criterion.

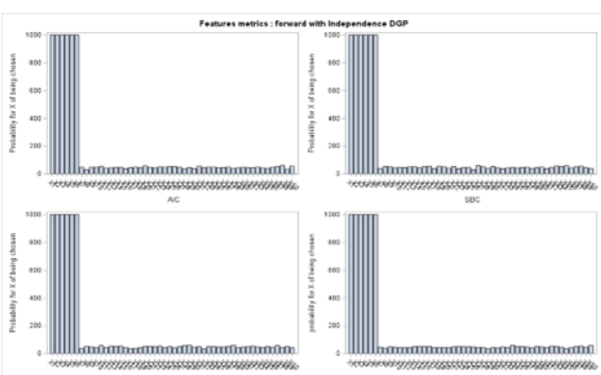


Figure 3: Features metrics : Forward in independence

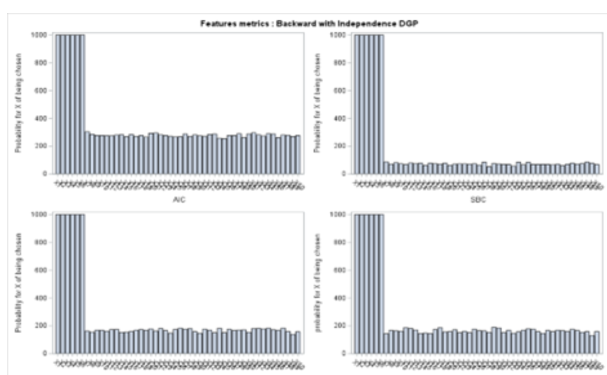


Figure 4: Features metrics : Backward in independence

This is in line with the feature metrics we found. The metric of a feature is the probability that it will be chosen by the selection algorithm. We can see that features X1 to X6 are always chosen with a probability of 100%, whether the method used is forward or backward, and whatever the stopping criteria. These metrics are higher with the backward method and this result is logical given that this method gave 100% overfitting for the AIC, AICC, and CP criteria as mentioned above.

4.1.2 Machine Learning : LASSO, LARS and Elasticnet

As part of our evaluation of machine learning selection algorithms, four criteria were tested: AIC and SBC for explanation, and K-Fold and Press for prediction. We used the LARS and LASSO algorithms, and our observations showed a close resemblance in the results generated by these two methods.

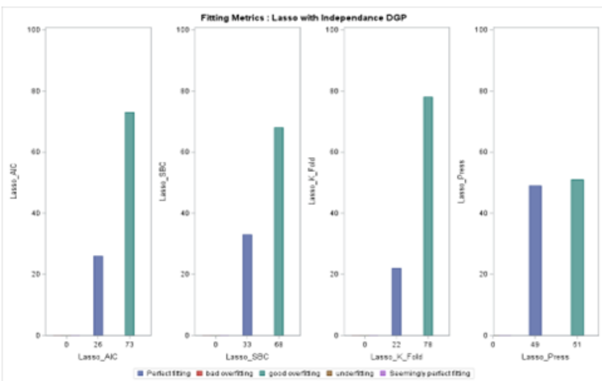


Figure 5: Fitting metrics : Lasso in independence

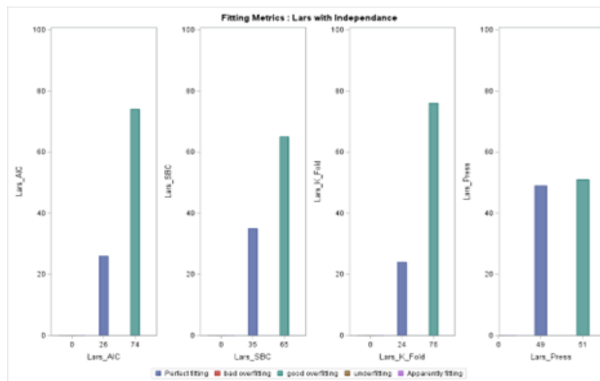


Figure 6: Fitting metrics : Lars in independence

These two methods showed relatively low percent of perfect fitting, ranging from 22% to 34% for the AIC, K-Fold and SBC criteria. For these three criteria, the good overfitting rate is quite high for LASSO and LARS : between 65 and 78%. There's no underfitting with these criteria. We get better results in terms of perfect fitting (49%) using PRESS on LASSO and LARS. For this criterion, there was no underfitting either, but 51% good overfitting. We can see that when selecting the best combination of selection and stop criteria, LASSO and LARS produce only perfect fitting and good overfitting.

The probability of choosing at least the good features is almost certain as it is made up of perfect fitting and predominant good overfitting with the AIC, SBC and K-Fold criteria. On the other hand, with the PRESS criterion, the right features are also all chosen, and the probability of other features being chosen in addition to the right ones already chosen is almost equal to perfect fitting rate. These single metrics are therefore coherent with the joinde metrics. In conclusion, PRESS is the best choice criterion for the LASSO and LARS methods.

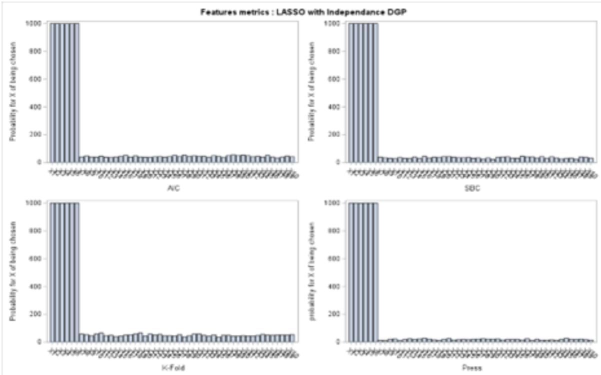


Figure 7: Features metrics : Lasso in independence

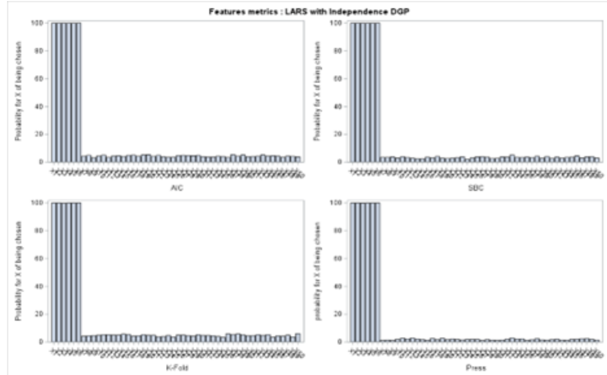


Figure 8: Features metrics : Lars in independence

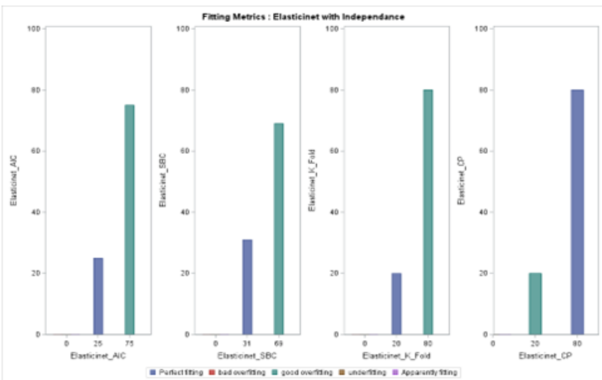


Figure 9: Fitting metrics : Elasticnet in independence

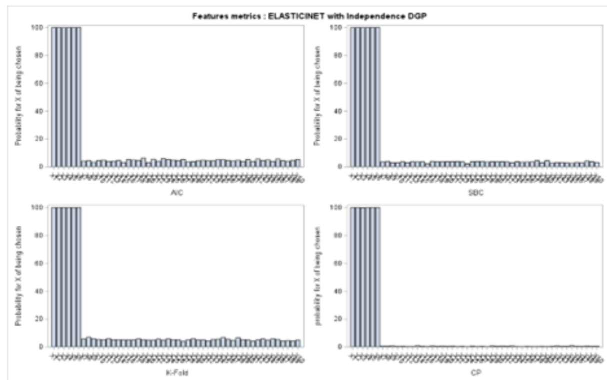


Figure 10: Features metrics : Elasticnet in independence

As for the Elastic Net, the results are similar to those of LASSO and LARS when applying the first three criteria, i.e. AIC, SBC and K-Fold. However, with the CP criterion, there is an explosion in perfect fitting, with a rate of 80

Like LASSO and LARS, Elastic Net gives a high probability of selecting at least all good features, with good overfitting (around 75%) dominating perfect fitting (25%) for AIC, SB and K-Fold. However, the situation changes completely when CP is used as a selection criterion. All the good features are chosen, but this time with perfect fitting (80%) far surpassing good overfitting (20%). This can be seen in the graph of single metrics, where the probability of other features being chosen apart from the right ones is low.

Overall, machine learning techniques are more effective and efficient than statistical learning approaches. In particular, Forward and Backward statistical learning methods show a high degree of good overfitting, which is a serious barrier to their capacity to perform perfect fitting. On the other hand, the Machine Learning methods Lars and Lasso provide far better results, especially when criteria of prediction such Press are applied (better perfect fitting). It is also important to underline a problem with our data generation process, i.e. the strong appearance of overfitting. A possible solution to this issue is to apply a second regression step after the initial selection by the algorithm. This additional step consists in removing ≪ irrelevant ≫ features, which should improve the probability of having a high perfect fitting rate and, at the same time, minimize overfitting. Of course, this approach is risky because of the possibility of deleting unexpected features that could still be statistically significant.

In brief, we can close this section by stating that most of these methods generally choose the right features over the incorrect ones.

4.2 Structural Break DGP

4.2.1 Statistical Learning : Forward and Backward

Forward and backward selection methods are built on the principle that the relations existing between features are uniform within the data set. But with the insertion of a structural break in the data, this hypothesis is not valid any more, and the efficiency of these methods may be impacted.

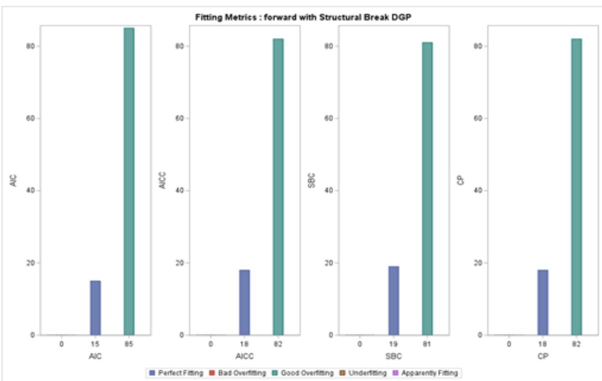


Figure 11: Fitting metrics : Forward in Break structural

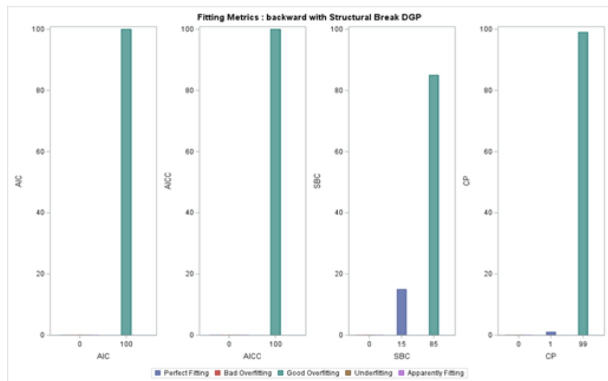


Figure 12: Fitting metrics : Backward in Break structural

For the DGP with structural break, we get 15-18% perfect fitting and 81-85% of good overfitting whatever the criterion for Forward selection method. But compared with Forward, Backward performs worse in terms of perfect fitting. There's no perfect fitting when applying AIC, AICC and CP. For these three, only 99-100% good overfitting is recorded. The criterion that stands out with the backward method is SBC (15% perfect fitting vs. 85% good overfitting).

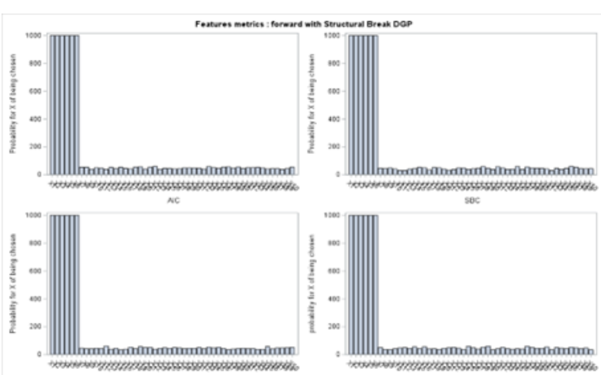


Figure 13: Features metrics : Forward in Break structural

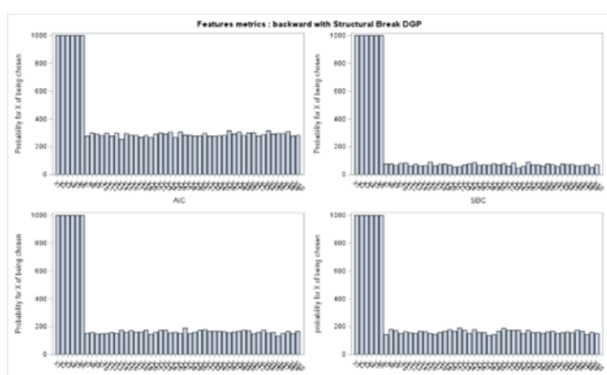


Figure 14: Features metrics : Backward in Break structural

These results show that the Forward method is better at capturing only the right features before and after the break than the Backward.

Comparative summary : the performance of the Forward and the Backward methods with the structural break are the same as for the DGP with white noise. They adjust to data split by the break, thanks to their iterative selection process, which is carried out variable by variable.

4.2.2 Machine Learning : LASSO, LARS and Elasticnet

Like statistical learning techniques, machine learning techniques are based on the hypothesis that intra-variable correlations are constant within the dataset.

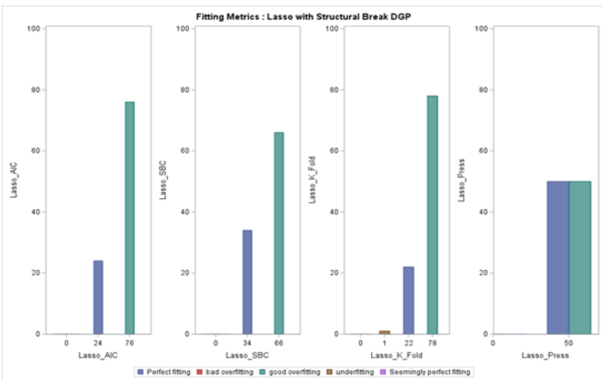


Figure 15: Fitting metrics : Lasso in Break structural

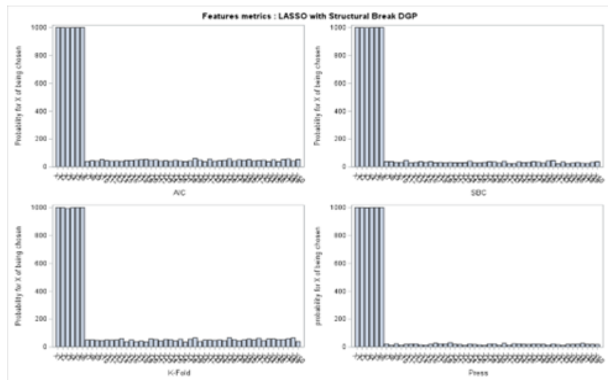


Figure 16: Features metrics : Lasso in Break structural

We obtain almost the same results as with DGP with white noise for these 3 methods. For LASSO, LARS and Elastic Net, we recorded between 19 and 34% perfect fitting, the rest being good overfitting with the AIC, SBC and AICC selection criteria. Only the PRESS criterion generates 50% perfect fitting and 50% overfitting for LASSO and LARS (see LARS graphs in appendix).

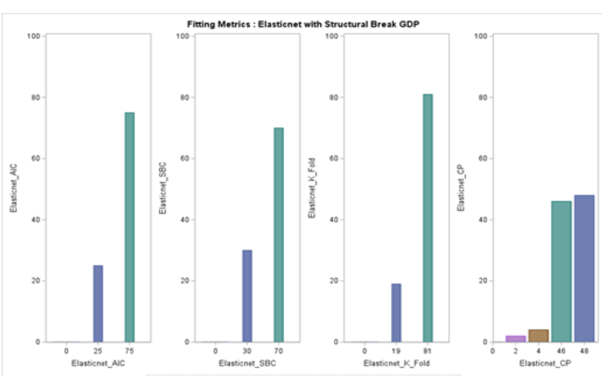


Figure 17: Fitting metrics : Elasticnet in Break structural

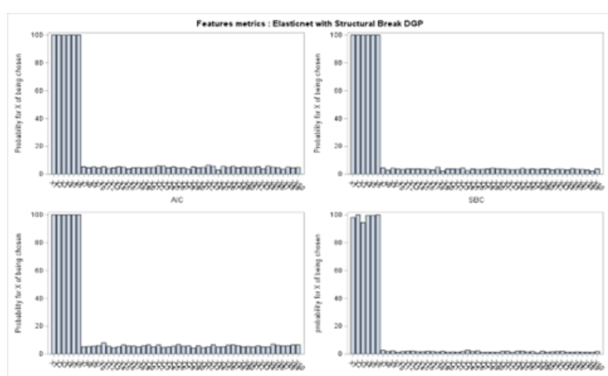


Figure 18: Features metrics : Elasticnet in Break structural

On the other hand, with Elastic Net - CP, we observed 48% perfect fitting and 46% good overfitting, resulting in a drop in performance in terms of perfect fitting compared to DGP with white noise (80% of perfect fitting and 20% of good overfitting). LASSO and LARS tend to be less affected by structural breaks. This is not really the case with Elastic Net - CP, with which we observe a drop in performance in terms of perfect fitting after the introduction of the structural break.

Comparative summary : In comparison with DGP with white noise, LASSO and LARS show consistent performance. However, the introduction of a structural break reduces the performance of the method-criterion pair : Elastic Net – CP.

4.3 Outliers DGP

The results are conditioned by the data generation process, where the optimum stopping criteria are selected from the choice criteria (AIC, AICC, SBC, CP, K-Fold and PRESS). This selection is made following a combination between each selection criterion and each stopping criterion, with the combination giving the best performance being chosen.

4.3.1 Statistical Learning : Forward and Backward

As expected, the presence of outliers in the generating data has a very significant impact on the Forward selection method. Indeed, the perfect fitting rate is between 4.8 and 7.4%, whatever the selection criterion used. The presence of outliers essentially causes overfitting, with bad overfitting slightly predominating (on average 32% vs. 30% good overfitting). We also note the presence of underfitting and apparently perfect fitting, each of which is on average twice as high as perfect fitting. These results are explained by the lower single probabilities of X1 and X3 being selected by the algorithm. In the DGP with independence (without outliers), the probability that the good features will all be selected is 100%. But with the introduction of outliers in the data, this metric drops to 82% on average for feature X1 and to around 42% for X3. This could be explained by the fact that these two features have the lowest coefficients : 1 for X1 and 0.6 for X3. The other coefficients, particularly X6, don't seem to be influenced by the outliers, probably because their coefficients are significant enough to neutralize the effect of the outliers

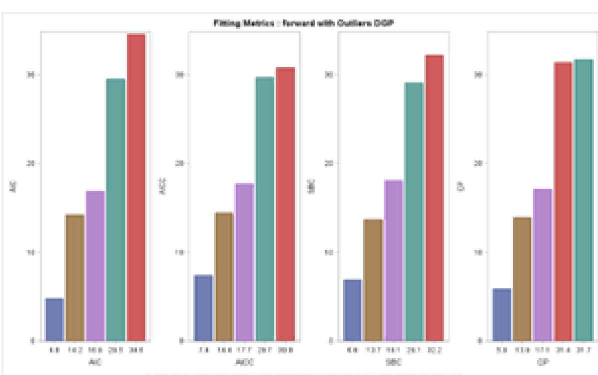


Figure 19: Fitting metrics : Forward in Outliers

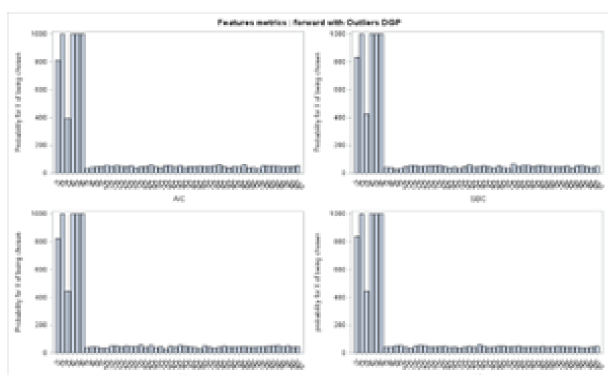


Figure 20: Features metrics : Forward in Outliers

The backward results don't show any improvement in performance, as outliers mainly generate overfitting and no perfect fitting with the AIC, AICC and CP selection criteria. Only the SBC results in 5% perfect fitting, and here again, good overfitting and bad overfitting are predominant. The explanation is the same, referring to the low coefficients associated with the X1 and X3 features. However, we could explain the better performance of Forward compared to Backward. By adding one variable at a time and evaluating its effect, the Forward method can better manage the perturbations caused by outliers, unlike the Backward method, which can exclude important features affected by outliers too early.

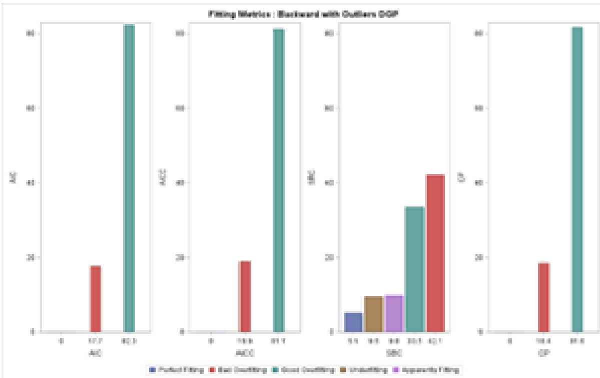


Figure 21: Fitting metrics : Backward in Outliers

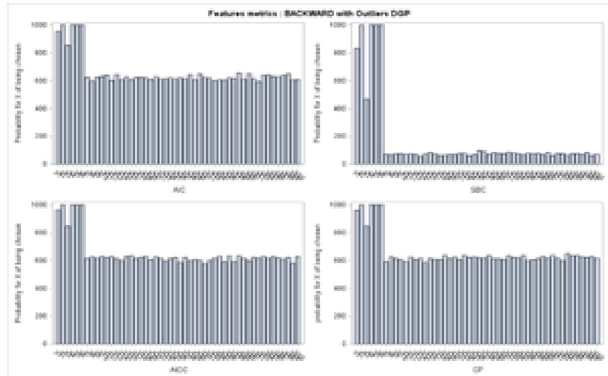


Figure 22: Features metrics : Backward in Outliers

4.3.2 Machine Learning : Lasso, Lars, Elasticnet

LASSO and LARS performance is also influenced by the presence of outliers. Using AIC, SBC and K-Fold, we get between 2% and 6% perfect fitting, which represents a reduction in performance compared with the case where there are no outliers in the data (the case of independence). Indeed, when there were no outliers in the data, the respective perfect fitting rates according to these criteria were on average 26%, 34% and 23% for LASSO and LARS. This decline in performance is even more significant with the PRESS criterion, where we go from 49% perfect fitting in the absence of outliers (independence) to 6% in the presence of outliers. If we take a closer look at single metrics, we see that all features are strongly impacted by the presence of outliers. Indeed, the individual probability of each feature Xi being chosen has fallen for the AIC, SBC and K-Fold criteria. The biggest decreases are for features with low coefficients, such as X1 and X3, which fall from 100% (absence of outliers) to an average of 38% and 14% respectively. This large reduction in single probabilities obviously leads to strong underfitting, averaging 68% for AIC, SBC and K-Fold.

With Press, only the probabilities of X1 and X3 being chosen have decreased. X1 and X3 go from 100% (absence of outliers) to 70% and 30% respectively (presence of outliers). This still leads to underfitting, but to a lesser degree (42%).

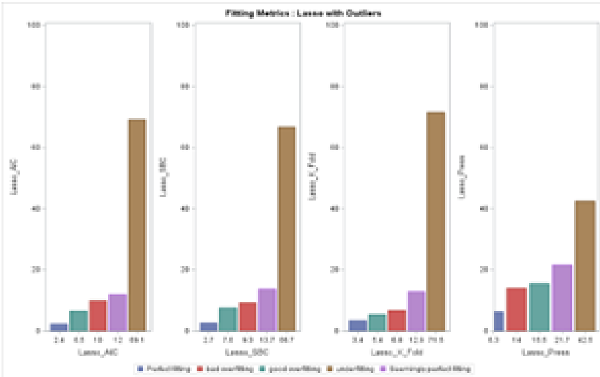


Figure 23: Fitting metrics : Lasso in Outliers

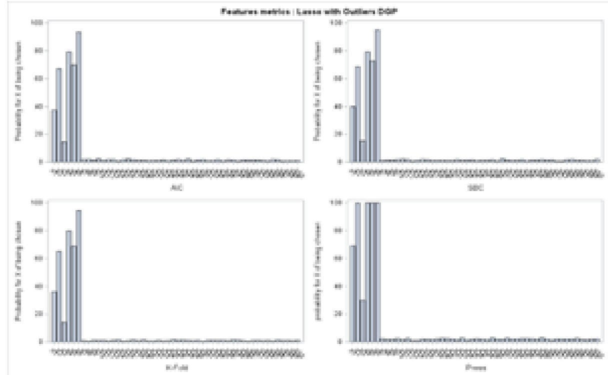


Figure 24: Features metrics : Lasso in Outliers

Concerning Elastic Net, we obtain practically the same results with AIC, SBC and K-Fold: very low perfect fitting (3-4% whatever the criterion), fairly high underfitting (22-34%) and mainly overfitting between 42-68%, with bad overfitting predominating. With CP, the results are worse, with 1% perfect fitting and 75% underfitting. Once again, this is due to the fact that outliers strongly impact features with low coefficients, and in the case of Elastic Net, X1 and X3 are affected by this decrease in probability. But the drop is greater with CP, where X1 and X3 fall from 100% (absence of outliers) to 30% and 10% respectively (presence of outliers).

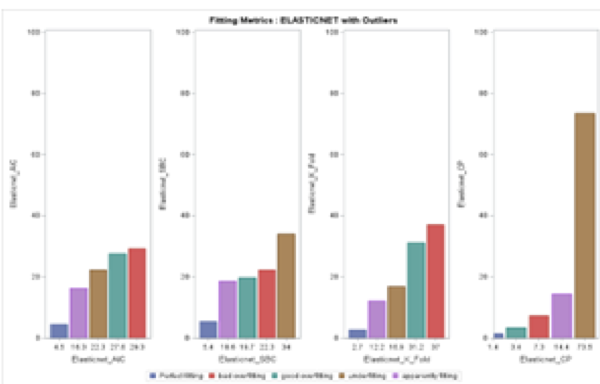


Figure 25: Fitting metrics : Elasticnet in Outliers

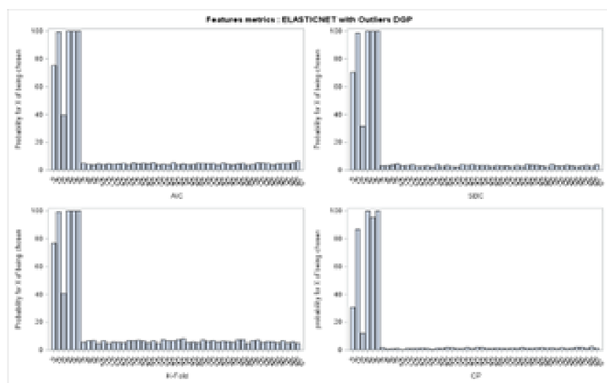


Figure 26: Features metrics : Elasticnet in Outliers

In summary, the gap in efficiency when using statistical learning and machine learning methods depends mainly on the choice of incorrect features. Statistical learning methods are more favorable to overfitting than LASSO and LARS. It's obvious that outliers lower the performance of each selection method, even if processes like Forward seem to give better results than Backward. LASSO and LARS give almost the same results. Elastic Net, on the other hand, gives more overfitting than LASSO and LARS. It is therefore imperative to take account of extreme values in the data. The simplest solution would be to remove them in order to clean up our dataset. However, this solution could be a mistake, as outliers are a component of the dataset and cancelling them could result in the loss of valuable information.

4.4 Intra-correlation : Correlation within the 6 first features

4.4.1 Statistical Learning algorithms: Forward and Backward

Theoretically, correlation among features can significantly impact the performance of statistical learning feature selection algorithms. In forward selection, variables are chosen one by one, by adding the one that significantly enhances the model performance. When our 6 predictors are correlated, it tends to always select these ones but it may also include some of the other variables if they improve the performance. This is called multicollinearity and can lead to overfitting of the model. Hence, we don't have underfitting in our results and we have a weak perfect fitting rate(average of 15.6%) compared to a high good overfitting rate (84.4%).

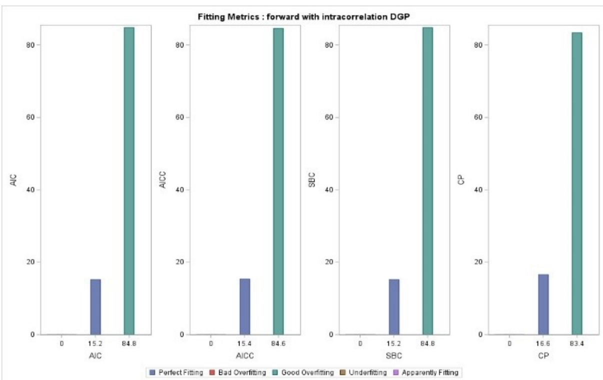


Figure 27: Fitting metrics : Forward in intra correlation

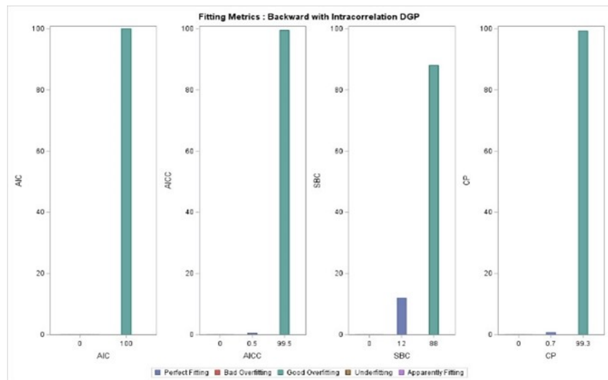


Figure 28: Fitting metrics : Backward in intra correlation

In backward selection, we start with all variables and we remove the less significant ones according to the selected criteria. Here again, multicollinearity will lead to keep the correlated predictors if they provide significant information in the model, together. This occurs because it's difficult to accurately determine the contribution of each feature. Our results are consistent with theory since we have even more overfitting (more than 99%) except with the SBC choose and stop criteria(12% perfect fitting).(see Forward/Backward feature metrics in appendix) In conclusion, despite the intra-correlation, the statistical learning algorithms performed similarly to when the variables were independent. The forward selection algorithm remains better than the backward in this case but they both got a strong tendency to overfitting. Note that all our results are contingent to the particular DGP and the selection methods and criteria.

4.4.2 Machine Learning algorithms: Lasso, Lars, Elasticnet

The impact of intra-correlation on Machine Learning's feature selection algorithms has yielded interesting observations and conclusions. In Lasso and Lars methods, we got similar results. We observed higher perfect fitting rates than with statistical algorithms but also some underfitting, particularly with AIC and K-fold choose criteria. We still observed some good overfitting in both methods and with any criteria (choose and stop). In terms of perfect fitting, the most efficient criteria combination in Lasso and Lars is CHOOSE=SBC STOP=CP, while the least is CHOOSE=PRESS STOP=SBC resulting 70% overfitting. In addition, when examining the feature metrics, we ob-

served a “masking effect”[9] , with the three first features being less selected than the three others. It can be explained by the fact that these 3 first are the highest correlated ones, leading Lasso/Lars to shrink the coefficients and select one of them(mainly X3) as representative(see Lars graphs in appendix). Nevertheless, we notice that changing the stop criteria affects the tradeoff between good overfitting and underfitting. For example, STOP=CP will provide high overfitting while STOP= SBC will result in good overfitting and underfitting.
Hence, testing different combinations can yield to different results.

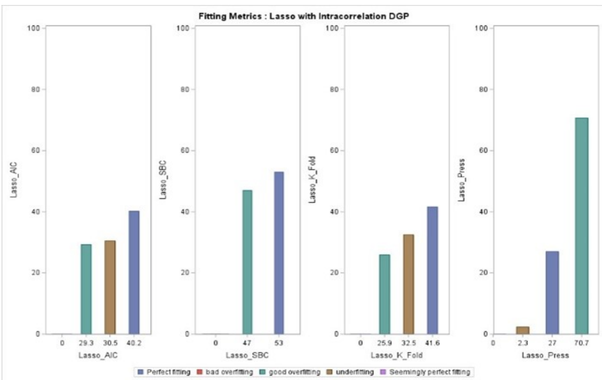


Figure 29: Fitting metrics : Lasso in intra correlation

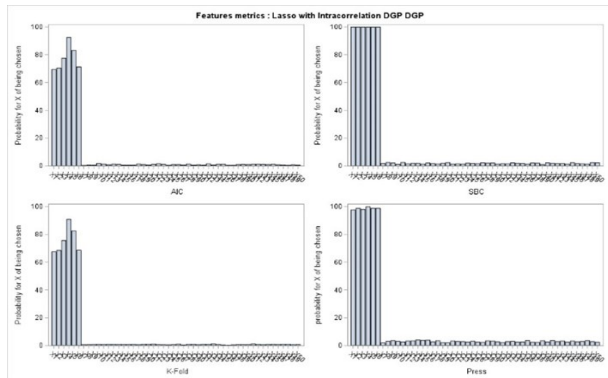


Figure 30: Features metrics : Lasso in intra correlation

Theoretically, one of the main advantages of the Elastic Net compared to the Lasso method is its ability to handle correlated predictor features more effectively. Our results confirmed the theory because the Elasticnet algorithm performed best with intra-correlation with an average of 60% chance of perfect fitting, with the combination CHOOSE=CP STOP=CP criteria as the best one(99.9% of perfect fitting and selecting the right features, but never the unexpected ones). However, the Elasticnet doesn’t fit as well with the K-fold choose criterion(33.1% of perfect fitting).

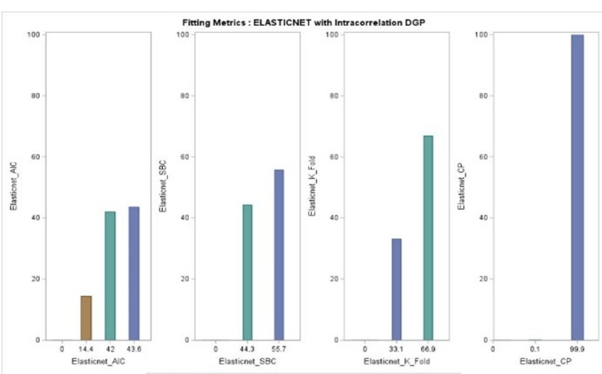


Figure 31: Fitting metrics : Elasticnet in intra correlation

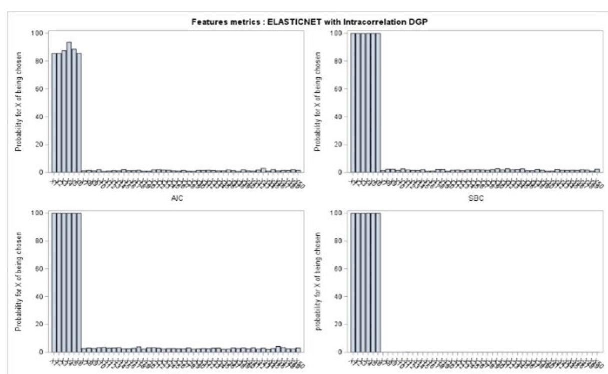


Figure 32: Features metrics : Elasticnet in intra correlation

In conclusion, the machine learning methods with intra-correlation provided us better performance than in the Independence DGP. We can also note that due to the correlation, some features will be ”masked” or, at least, their coefficient will be biased. Moreover, all our results are conditionally to the DGP, the noise effect and the coefficient vector.

4.5 Inter-correlation : Correlation between all the variables

4.5.1 Statistical Learning algorithms : Forward and Backward

The correlation among all variables can significantly impact the performance of statistical learning feature selection algorithms. On one hand, in the forward method, the presence of multicollinearity can lead to overfitting. When all variables are correlated, the forward algorithm tends to add the first six highly correlated features, but it may also include others if they provide additional information to the model. Thus, observing our fitting metrics, we found a 31% chance of perfect fitting and 69% chance of good overfitting on average. The feature metrics reveal that predictor features are consistently selected, with a high probability of selecting unexpected variables.

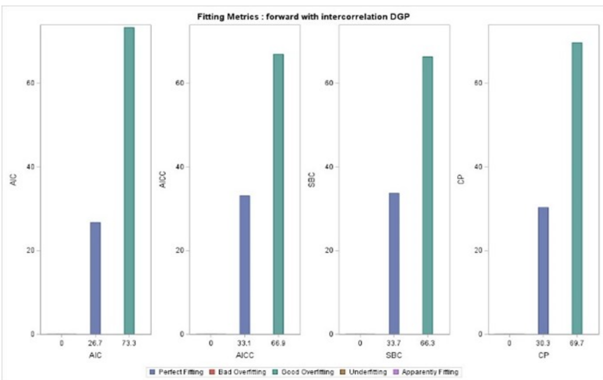


Figure 33: Fitting metrics : Forward in inter correlation

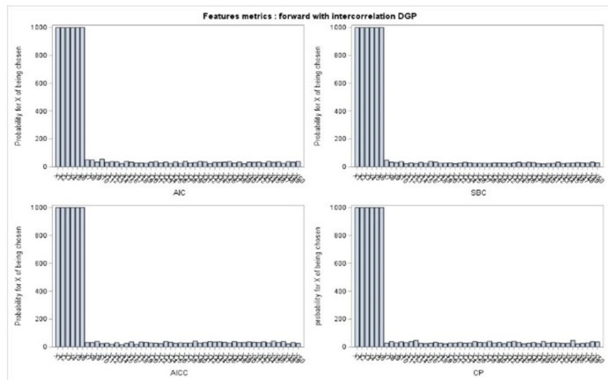


Figure 34: Features metrics : Forward in intra correlation

On the other hand, the performance of the backward method remains unchanged, with more than a 99% chance of overfitting, except for CHOOSE=SBC, which has a 12.4% chance of achieving perfect fitting. Upon observing the feature metrics, we found a higher marginal probability of selecting a unexpected variable (more than 25% for CHOOSE=AIC!). Particularly, the AICC choose criterion exhibits a smaller probability of selecting other variables compared to the AIC criterion, as AICC is more suitable when there are many parameters and fewer observations.

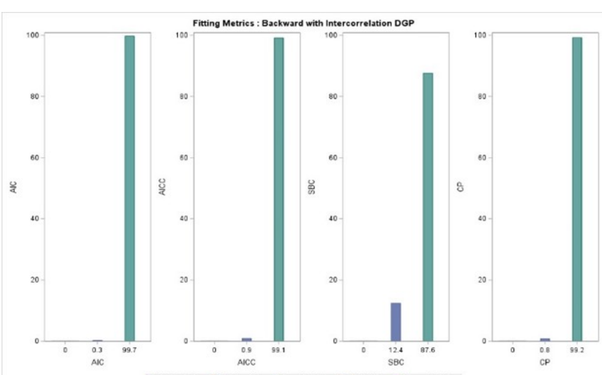


Figure 35: Fitting metrics : Backward in inter correlation

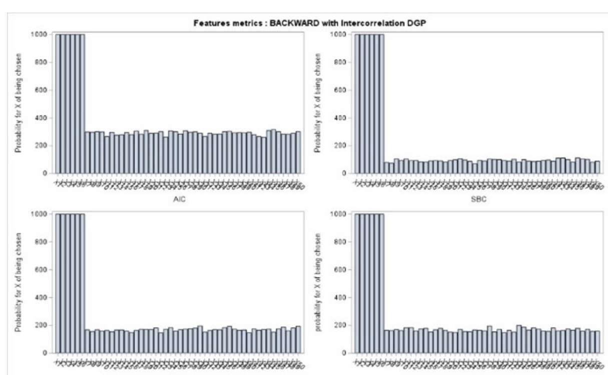


Figure 36: Features metrics : Backward in intra correlation

Hence, we can conclude that correlated variables can lead to unstable and unreliable parameter estimates in backward selection, while unexpected variables are more easily identified and removed in forward selection. Once again, the forward method outperformed the backward method, as observed in the Independence DGP. We also observed that, unlike the intra-correlation DGP, the higher the correlation coefficients, the higher the chance of achieving perfect fitting. It's important to note that these assumptions depend on the specific DGP and the choice of coefficients and noise effect.

4.5.2 Machine Learning algorithms: Lasso, Lars, Elasticnet

Theoretically, we can expect some overfitting or underfitting due to multicollinearity. Our fitting metrics indicate that Lasso and Lars methods share similar performance characteristics in terms of overfitting and perfect fitting rates. According to the chosen criteria and stopping rules, we observed an average of 48% perfect fitting in both Lasso and Lars, with no underfitting (depending on the stopping criteria). The combination CHOOSE=PRESS for Lasso/Lars proves to be the most effective, with an almost 60% chance of achieving perfect fitting. Additionally, upon observing the feature metrics, we noticed that the expected features are consistently selected, with a decrease in the marginal probability of choosing an unexpected variable, attributed to the smaller and decreasing correlation among them.(see Lasso graphs in appendix)

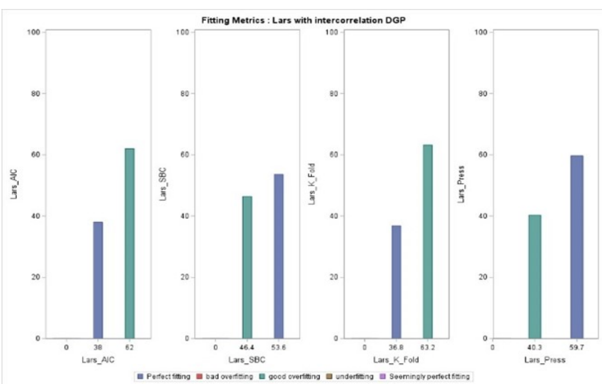


Figure 37: Fitting metrics : Lars in inter correlation

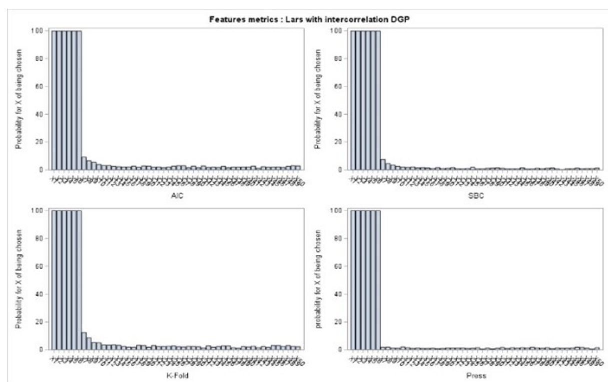


Figure 38: Features metrics : Lars in intra correlation

At the same time, the results of Elastic Net demonstrate better performance in terms of perfect fitting probabilities (averaging 62%), particularly with the combination CHOOSE=CP STOP=CV achieving 85.4% perfect fitting. The feature metrics also indicate a decrease in the marginal probability of choosing an unexpected variable due to the same reason

.

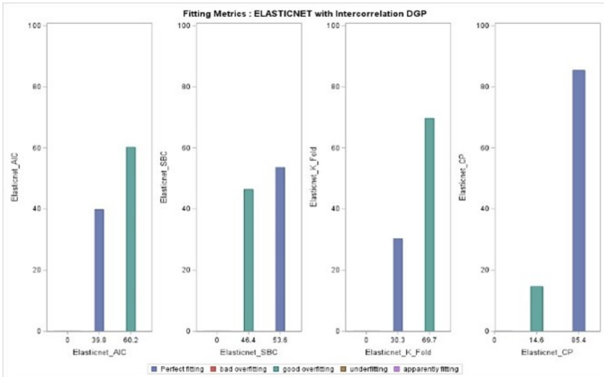


Figure 39: Fitting metrics : Elasticnet in inter correlation

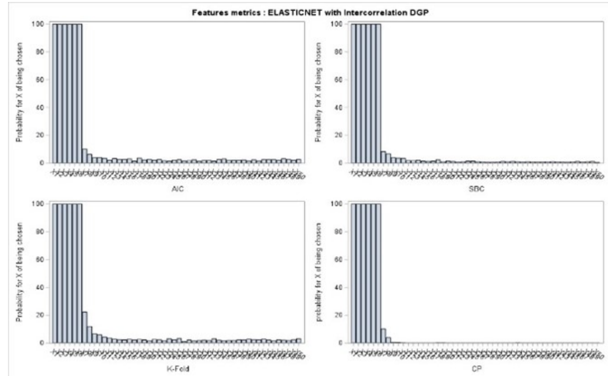


Figure 40: Features metrics : Elasticnet in intra correlation

In summary, the efficacy of mitigating correlation effects hinges on choosing appropriate criteria and is influenced by correlation strength, coefficient vector composition, and noise levels, with higher correlation coefficients generally enhancing model fitting accuracy.

5 Empirical Analysis

5.1 Data Analysis

This part describes how our selection methods work with real data. We have a dataset of 43,020 observations and 12 features. These are composed of a dependent variable (nbdoct) which is the number of medical visits as a function of 11 independent variables such as age, income, whether a person has arthritis or not, whether they have diabetes or not, and so on. By first examining the data set, we can already get an idea of the shape of the variable distribution with this tab

Le Système SAS

La procédure MEANS

Variable	Libellé	N	Moyenne	Médiane	Minimum	Maximum	Erreur type	Skewness	Kurtosis
nbdoct	nbdoct	43020	4.9095537	3.0000000	0	365.0000000	0.0477572	12.7038008	301.3790073
female	female	43020	0.5938633	1.0000000	0	1.0000000	0.0023678	-0.3822625	-1.8539616
age	age	43020	50.7717806	50.0000000	18.0000000	85.0000000	0.0826130	0.1297168	-0.7766597
income	income	43020	68741.80	50000.00	0	300000.00	292.1992392	1.7279818	3.3747534
asthma	asthma	43020	0.0874709	0	0	1.0000000	0.0013622	2.9204140	6.5291216
diabetes	diabetes	43020	0.0778243	0	0	1.0000000	0.0012916	3.1519117	7.9349161
hbpressure	hbpressure	43020	0.2998373	0	0	1.0000000	0.0022091	0.8737476	-1.2366226
heart	heart	43020	0.0855649	0	0	1.0000000	0.0013486	2.9633132	6.7815405
arthritis	arthritis	43020	0.2589958	0	0	1.0000000	0.0021122	1.1003048	-0.7893660
partner	partner	43020	0.0584844	0	0	1.0000000	0.0011314	3.7632000	12.1622395
bornus	bornus	43020	0.7694561	1.0000000	0	1.0000000	0.0020307	-1.2795709	-0.3627153
mental	mental	43020	0.0385402	0	0	1.0000000	0.000928095	4.7946441	20.9895880

Table 3

Zero skewness means that the distribution is well balanced on both sides. A skewness very far from zero shows that the dispersion is either on the left (negative skewness) or on the right (positive skewness).

The higher the kurtosis, the more outliers. For instance, for these still non-log-linearized data, features such as mental, partner, heart, diabetes have high kurtosis and skewness, hence a strong presence of outliers for these features. So, as we have seen with simulated data, the existence of outliers is problematic. By calculating the variance-covariance matrix, we can note that some variables are quite correlated, such as "age" and "hbpressure" (39.7%) or "age" and "arthritis" (38%) or "hbpressure" and "arthritis" with around 25% correlation. These fairly high correlations could cause problems for the methods used to fit the model, resulting in an underfitting.

Coefficients de corrélation de Pearson, N = 43020
Proba > |r| sous H0: Rho=0

	nbdoct	female	age	income	asthma	diabetes	hbpressure	heart	arthritis	partner	bornus	mental
nbdoct	1.00000	0.06529	0.09585	-0.04587	0.09250	0.10789	0.11983	0.12791	0.15896	-0.01152	0.07209	0.10026
nbdoct		<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	0.0169	<.0001	<.0001
female	0.06529	1.00000	0.04155	-0.10144	0.06656	-0.03114	-0.01325	-0.03486	0.10169	-0.00185	0.00078	0.03353
female	<.0001		<.0001	<.0001	<.0001	<.0001	0.0060	<.0001	<.0001	0.7016	0.8716	<.0001
age	0.09585	0.04155	1.00000	-0.08545	0.00136	0.17603	0.39680	0.28088	0.38002	-0.12954	0.14950	-0.02444
age	<.0001	<.0001		<.0001	0.7775	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001
income	-0.04587	-0.10144	-0.08545	1.00000	-0.03229	-0.09971	-0.10652	-0.07146	-0.11016	-0.01487	0.10994	-0.11291
income	<.0001	<.0001	<.0001		<.0001	<.0001	<.0001	<.0001	<.0001	0.0020	<.0001	<.0001
asthma	0.09250	0.06656	0.00136	-0.03229	1.00000	0.02922	0.05597	0.05530	0.10262	-0.00038	0.08645	0.07179
asthma	<.0001	<.0001	0.7775	<.0001		<.0001	<.0001	<.0001	<.0001	0.9376	<.0001	<.0001
diabetes	0.10789	-0.03114	0.17603	-0.09971	0.02922	1.00000	0.22160	0.15495	0.11149	-0.01916	-0.00332	0.05137
diabetes	<.0001	<.0001	<.0001	<.0001	<.0001		<.0001	<.0001	<.0001	<.0001	0.4904	<.0001
hbpressure	0.11983	-0.01325	0.39680	-0.10652	0.05597	0.22160	1.00000	0.23874	0.25549	-0.04138	0.08128	0.04398
hbpressure	<.0001	0.0060	<.0001	<.0001	<.0001	<.0001		<.0001	<.0001	<.0001	<.0001	<.0001
heart	0.12791	-0.03486	0.28088	-0.07146	0.05530	0.15495	0.23874	1.00000	0.17597	-0.03233	0.05932	0.04625
heart	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001		<.0001	<.0001	<.0001	<.0001
arthritis	0.15896	0.10169	0.38002	-0.11016	0.10262	0.11149	0.25549	0.17597	1.00000	-0.04492	0.13943	0.08093
arthritis	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001		<.0001	<.0001	<.0001
partner	-0.01152	-0.00185	-0.12954	-0.01487	-0.00038	-0.01916	-0.04138	-0.03233	-0.04492	1.00000	-0.01551	0.00928
partner	0.0169	0.7016	<.0001	0.0020	0.9376	<.0001	<.0001	<.0001	<.0001		0.0013	0.0543
bornus	0.07209	0.00078	0.14950	0.10994	0.08645	-0.00332	0.08128	0.05932	0.13943	-0.01551	1.00000	-0.01914
bornus	<.0001	0.8716	<.0001	<.0001	<.0001	0.4904	<.0001	<.0001	<.0001	0.0013		<.0001
mental	0.10026	0.03353	-0.02444	-0.11291	0.07179	0.05137	0.04398	0.04625	0.08093	0.00928	-0.01914	1.00000
mental	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	0.0543	<.0001	

Table 4

5.2 Results

The graphs show the percentages of selection of the variables in 2 cases. In the first case, we used the original data with both Statistical Learning and Machine learning methods, particularly the stepwise and the Lasso.

Due to the presence of outliers and correlation, we got different results. With Stepwise method, we have underfitting because it selects only 7 variables perfectly and the others like income or partner are less or never selected. With the Lasso method, we have better marginal probabilities of selecting the variables (¿40%) but they are not the same variables as the Stepwise method. That's the masking effect previously seen. The Lasso seems being better than the stepwise but it's contingent to the data structure(the noise effects, correlation. . .). Theoretically, we saw that even though these processes can tend to overfitting or underfitting, they still have a tendency to select the good features.

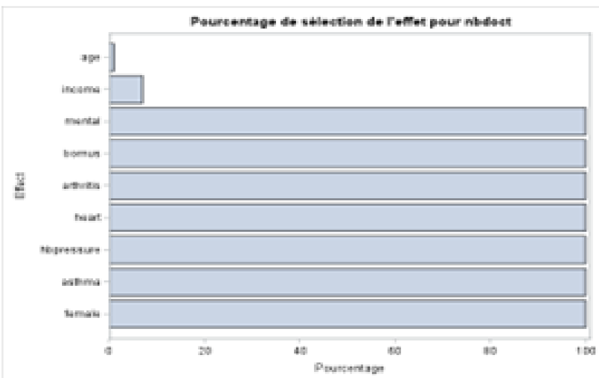


Figure 41: Stepwise

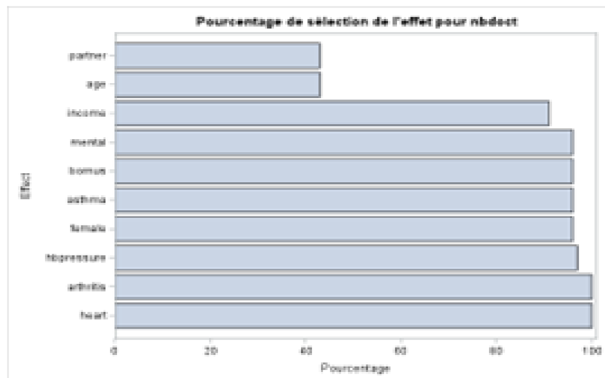


Figure 42: Lasso

In the second case, we used the original data corrected with log-linearisation on the variables nbdoct and income. We chose to apply the log on these variables dur to their positive and high skewnesses. We also tested both Statistical Learning and Machine learning methods, particularly the stepwise and the Lasso.

Due to the presence of outliers and correlation, we got once again different results. With Stepwise method, we have a fewer underfitting with only one variable (partner) poorly selected . With the Lasso method, we have better marginal probabilities of selecting the variables but we still have some underfitting. By correcting the data, we have a good amount of variables selected but it also depends on the criteria and selected algorithms.

Hence, it's important to consider the link between theoretical assumptions and the actual data before choosing the right selection model.

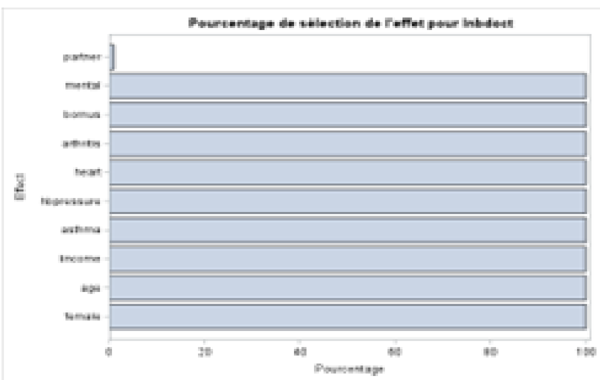


Figure 43: Stepwise

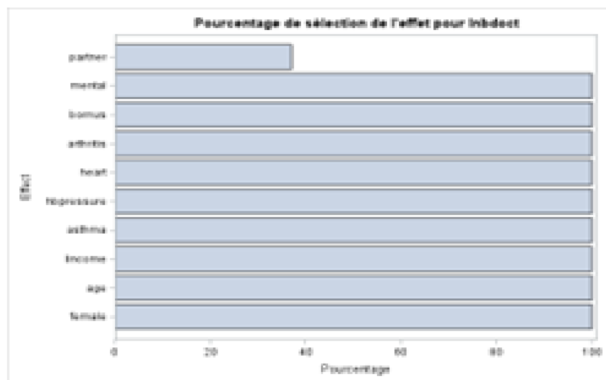


Figure 44: Lasso

6 Conclusion

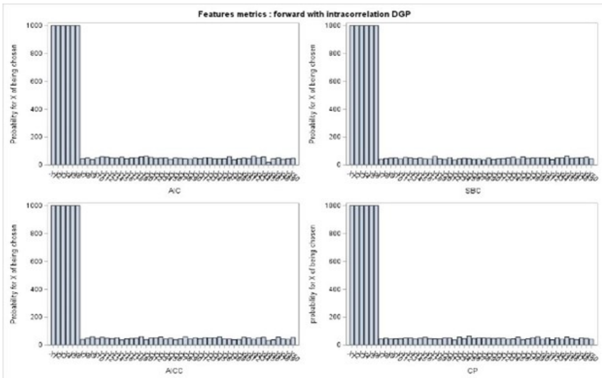
This paper aimed to conduct an in-depth investigation of feature selection methodologies within the realms of machine learning and statistical learning, highlighting the distinctions and interactions between these two approaches, particularly in the context of selecting features. It was observed that despite sharing a common objective, the paths they follow diverge significantly due to their foundational principles—one being inference-driven and the other not strictly adhering to the statistical laws governing data distribution. A key finding of our research is that each algorithm comes with its own set of advantages and drawbacks. Algorithms from the machine learning domain tend to minimize features that have a minor yet significant effect on the outcome variable, whereas those from statistical learning are prone to overfitting issues. However, statistical learning models depend heavily on strong assumptions about data distribution, which can be a limitation as the validity of their outcomes hinges on these assumptions. In contrast, machine learning models exhibit high sensitivity to noise and the values of predictor variables.

By altering the foundational assumptions of our dataset, we were able to pinpoint instances where these methodologies faltered in selecting the appropriate model. This underscores the importance of preliminary dataset analysis to ascertain the potential performance of algorithms, identifying which might underperform or excel. Furthermore, the stochastic nature of these methods means that alterations to the dataset can yield varying outcomes. We noted that while some techniques might fail on a single dataset, applying them across multiple datasets from the same distribution generally led to the selection of more relevant features. Techniques such as bootstrapping enhanced the robustness of our results. The choice of criteria, whether for explanatory or predictive purposes, proved critical. Ultimately, the adoption of a theoretical perspective and intuitive understanding of the algorithms is crucial.

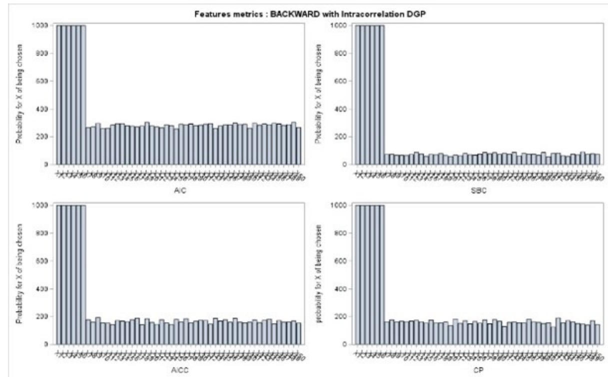
Our study acknowledges certain limitations. For instance, the Data Generating Processes (DGP) were simulated based on an ideal multivariate normal distribution, which is rarely encountered in real-world data that often exhibits correlations, outliers, and structural breaks simultaneously.

Looking forward, this paper has opened avenues for further research in not just feature selection but also in the estimation of feature coefficients. Investigating the performance of these algorithms in estimating feature coefficients, as well as examining the efficacy of combining multiple feature selection techniques, could potentially lead to superior outcomes than employing a singular technique.

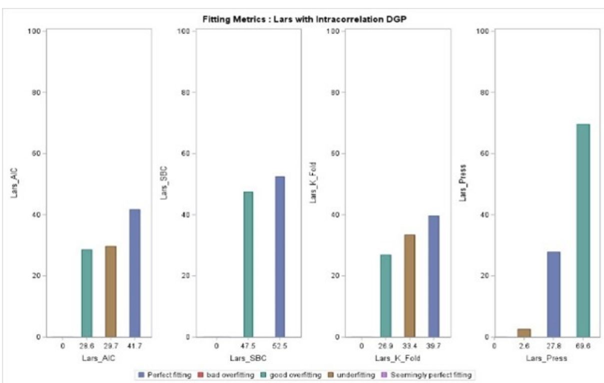
7 Appendix



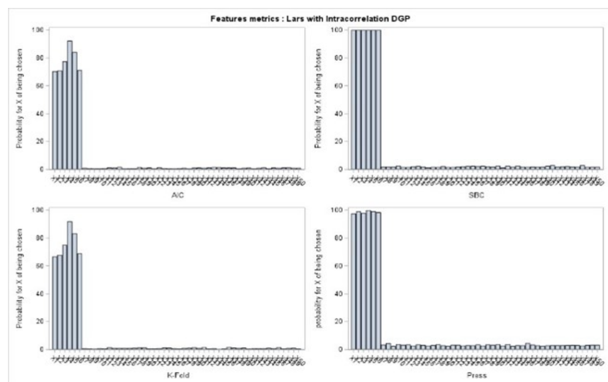
Features metrics : forward with Intracorrelation DGP
AIC
SBC
AICC
CP



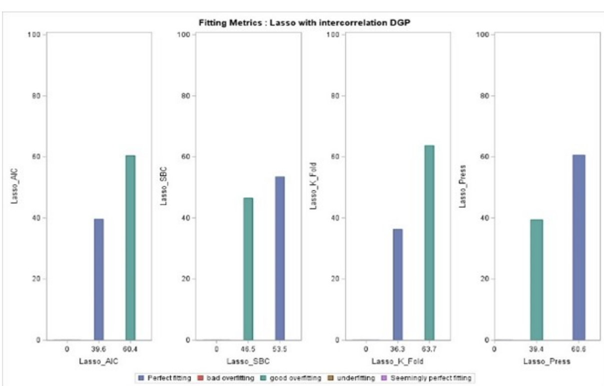
Features metrics : BACKWARD with Intracorrelation DGP
AIC
SBC
AICC
CP



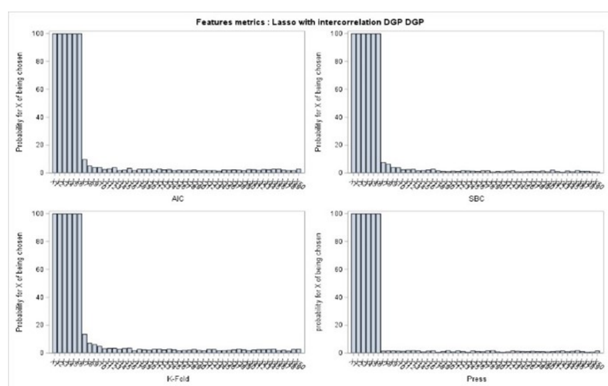
Fitting Metrics : Lars with Intracorrelation DGP
0
28.6
29.7
41.7
Lars_AIC
0
47.5
52.5
Lars_SBC
0
26.9
33.4
39.7
Lars_K_Fold
0
2.6
27.8
69.6
Lars_Press
Perfect fitting
bad overfitting
good overfitting
underfitting
Seemingly perfect fitting



Features metrics : Lars with Intracorrelation DGP
AIC
SBC
K-Fold
Press



Fitting Metrics : Lasso with intercorrelation DGP
0
39.6
60.4
Lasso_AIC
0
46.5
53.5
Lasso_SBC
0
36.3
63.7
Lasso_K_Fold
0
39.4
60.6
Lasso_Press
Perfect fitting
bad overfitting
good overfitting
underfitting
Seemingly perfect fitting



Features metrics : Lasso with intercorrelation DGP DGP
AIC
SBC
K-Fold
Press

8 Bibliography

References

[1] Hirotogu Akaike. “Information Theory and an Extension of the Maximum Likelihood Principle”. en. In: *Selected Papers of Hirotugu Akaike*. Ed. by Emanuel Parzen, Kunio Tanabe, and Genshiro Kitagawa. Series Title: Springer Series in Statistics. New York, NY: Springer New York, 1998, pp. 199–213. ISBN: 978-1-4612-7248-9 978-1-4612-1694-0. DOI: 10.1007/978-1-4612-1694-0_15. URL: http://link.springer.com/10.1007/978-1-4612-1694-0_15 (visited on 01/21/2023).

[2] Hirotugu Akaike. “A new look at the statistical model identification”. In: *IEEE transactions on automatic control* 19.6 (1974), pp. 716–723.

[3] David M Allen. “The relationship between variable selection and data agumentation and a method for prediction”. In: *technometrics* 16.1 (1974), pp. 125–127.

[4] Efron Bradley et al. “Least angle regression”. In: *The Annals of statistics* 32.2 (2004), pp. 407–499.

[5] Prabir Burman. “A comparative study of ordinary cross-validation, v-fold cross-validation and the repeated learning-testing methods”. In: *Biometrika* 76.3 (1989), pp. 503–514.

[6] R. A. Fisher. “On a distribution yielding the error functions of several well known statistics”. In: (1924).

[7] Jerome Friedman et al. “[Consistency in Boosting]: Discussion”. In: *The Annals of Statistics* 32.1 (2004), pp. 102–107.

[8] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY: Springer, 2009. ISBN: 978-0-387-84857-0 978-0-387-84858-7. DOI: 10.1007/978-0-387-84858-7. URL: http://link.springer.com/10.1007/978-0-387-84858-7 (visited on 01/21/2023).

[9] Trevor Hastie et al. *The elements of statistical learning: data mining, inference, and prediction*. Vol. 2. Springer, 2009.

[10] Arthur E Hoerl and Robert W Kennard. “Ridge regression: Biased estimation for nonorthogonal problems”. In: *Technometrics* 12.1 (1970), pp. 55–67.

[11] Colin L Mallows. “Some comments on Cp”. In: *Technometrics* 42.1 (2000), pp. 87–94.

[12] David B Montgomery and Donald G Morrison. “A Note on Adjusting— mathrmR 2”. In: *The Journal of Finance* 28.4 (1973), pp. 1009–1013.

[13] J. Neyman and E. S. Pearson. “On the Use and Interpretation of Certain Test Criteria for Purposes of Statistical Inference: Part I”. en. In: *Biometrika* 20A.1/2 (July 1928), p. 175. ISSN: 00063444. DOI: 10.2307/2331945. URL: https://www.jstor.org/stable/2331945?origin=crossref (visited on 01/21/2023).

[14] Takamitsu Sawa. "Information Criteria for Discriminating Among Alternative Regression Models". In: *Econometrica* 46.6 (1978). Publisher: [Wiley, Econometric Society], pp. 1273–1291. ISSN: 0012-9682. DOI: 10.2307/1913828. URL: https://www.jstor.org/stable/1913828 (visited on 01/21/2023).

[15] Gideon Schwarz. "Estimating the Dimension of a Model". In: *The Annals of Statistics* 6.2 (1978), pp. 461–464. URL: http://www.jstor.org/stable/2958889.

[16] Mary L Thompson. "Selection of variables in multiple regression: Part I. A review and evaluation". In: *International Statistical Review/Revue Internationale de Statistique* (1978), pp. 1–19.

[17] Robert Tibshirani. "Regression Shrinkage and Selection via the Lasso". In: (1994).

[18] Vladimir Vapnik. "The Nature of Statistical learning theory". In: (2010).

[19] Fengrong Wei, Jian Huang, and Hongzhe Li. "Variable selection and estimation in high-dimensional varying-coefficient models". In: *Statistica Sinica* 21.4 (2011), p. 1515.

[20] Hui Zou and Trevor Hastie. "Regularization and variable selection via the elastic net". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 67.2 (2005), pp. 301–320.